Additive identity: $\phi_{\mathbf{s}}(0) = 0$ for all $\mathbf{s} \in S$, so $0 \in I(S)$.

Additive inverses: $f(\mathbf{x}) \in I(S)$ implies $\phi_{\mathbf{s}}(-f(\mathbf{x})) = -\phi_{\mathbf{s}}(f(\mathbf{x})) = -f(\mathbf{s}) = -0 = 0$ for all $\mathbf{s} \in S$, so $-f(\mathbf{x}) \in I(S)$.

Multiplicative property: Let $f(\mathbf{x}) \in I(S)$ and $h(\mathbf{x}) \in F[\mathbf{x}]$. Then $\phi_{\mathbf{s}}(h(\mathbf{x})f(\mathbf{x})) = \phi_{\mathbf{s}}(h(\mathbf{x}))\phi_{\mathbf{s}}(f(\mathbf{x})) = h(\mathbf{s})f(\mathbf{s}) = h(\mathbf{s})(0) = 0$ for all $\mathbf{s} \in S$, so $h(\mathbf{x})f(\mathbf{x}) \in I(S)$.

33. By Definition 28.1, $V(I(S))$ consists of all common zeros of elements of $I(S)$. By definition of $I(S)$ in Exercise 32, every $f(\mathbf{x}) \in I(S)$ has every $\mathbf{s} \in S$ as a zero, so $\mathbf{s} \in V(I(S))$ for all $\mathbf{s} \in S$. Thus $S \subseteq V(I(S))$.

34. Let $n = 1, F = \mathbb{Q}$ and $S = \mathbb{Z} \subseteq \mathbb{Q}$ in Exercise 32. Then $I(S) = \{0\}$, for a nonzero element of $\mathbb{Q}[x]$ can have only a finite number of zeros in $\mathbb{Q}$. However, $V(I(S)) = V(\{0\}) = \mathbb{Q} \neq \mathbb{Z}$.

35. Let $f(\mathbf{x}) \in N$. Now $V(N)$ consists of all $\mathbf{s} \in F^n$ such that $h(\mathbf{s}) = 0$ for all $h(\mathbf{x}) \in N$. Thus $\mathbf{s} \in V(N)$ implies that $f(\mathbf{s}) = 0$ because $f(\mathbf{x})$ is an example of an $h(\mathbf{x}) \in N$. Thus $f(\mathbf{s}) = 0$ for all $\mathbf{s} \in V(N)$. By definition of $I(S)$ in Exercise 32, this means that $f(\mathbf{x}) \in I(V(N))$. Thus $N \subseteq I(V(N))$.

36. Let $N = \langle x^2, y^2 \rangle$. Then $V(N) = \{(0,0)\}$, and $I(V(N)) = \langle x, y \rangle \neq N$.

29. Introduction to Extension Fields

1. Let $\alpha = 1 + \sqrt{2}$. Then $(\alpha - 1)^2 = 2$ so $\alpha^2 - 2\alpha - 1 = 0$. Thus α is a zero of $x^2 - 2x - 1$ in $\mathbb{Q}[x]$.

2. Let $\alpha = \sqrt{2}+\sqrt{3}$. Then $\alpha^2 = 2+2\sqrt{6}+3$ so $\alpha^2-5 = 2\sqrt{6}$. Squaring again, we obtain $\alpha^4-10\alpha^2+1 = 0$, so α is a zero of $x^4 - 10x^2 + 1$ in $\mathbb{Q}[x]$.

3. Let $\alpha = 1 + i$. Then $(\alpha - 1)^2 = -1$, so $\alpha^2 - 2\alpha + 2 = 0$. Thus α is a zero of $x^2 - 2x + 2$ in $\mathbb{Q}[x]$

4. Let $\alpha = \sqrt{1 + \sqrt[3]{2}}$. Then $\alpha^2 = 1 + \sqrt[3]{2}$ so $\alpha^2 - 1 = \sqrt[3]{2}$. Cubing, we obtain $\alpha^6 - 3\alpha^4 + 3\alpha^2 - 3 = 0$, so α is a zero of $x^6 - 3x^4 + 3x^2 - 3$ in $\mathbb{Q}[x]$.

5. Let $\alpha = \sqrt{\sqrt[3]{2} - i}$. Then $\alpha^2 + i = \sqrt[3]{2}$. Cubing, we obtain $\alpha^6 + 3\alpha^4 i - 3\alpha^2 - i = 2$, so $\alpha^6 - 3\alpha^2 - 2 = (1 - 3\alpha^4)i$. Squaring, we obtain $\alpha^{12} - 6\alpha^8 - 4\alpha^6 + 9\alpha^4 + 12\alpha^2 + 4 = -1 + 6\alpha^4 - 9\alpha^8$. Thus $\alpha^{12} + 3\alpha^8 - 4\alpha^6 + 3\alpha^4 + 12\alpha^2 + 5 = 0$, so α is a zero of $x^{12} + 3x^8 - 4x^6 + 3x^4 + 12x^2 + 5$ in $\mathbb{Q}[x]$.

6. Let $\alpha = \sqrt{3 - \sqrt{6}}$. Then $\alpha^2 - 3 = -\sqrt{6}$. Squaring again, we obtain $\alpha^4 - 6\alpha^2 + 3 = 0$, so α is a zero of $f(x) = x^4 - 6x^2 + 3$ in $\mathbb{Q}[x]$. Now $f(x)$ is monic and is irreducible by the Eisenstein condition with $p = 3$. Thus $\deg(\alpha, \mathbb{Q}) = 4$ and $\text{irr}(\alpha, \mathbb{Q}) = f(x)$.

7. Let $\alpha = \sqrt{\frac{1}{3} + \sqrt{7}}$. Then $\alpha^2 - \frac{1}{3} = \sqrt{7}$. Squaring again, we obtain $\alpha^4 - \frac{2}{3}\alpha^2 - \frac{62}{9} = 0$, or $9\alpha^4 - 6\alpha^2 - 62 = 0$. Let $f(x) = 9x^4 - 6x^2 - 62$. Then $f(x)$ is irreducible by the Eisenstein condition with $p = 2$. Thus $\deg(\alpha, \mathbb{Q}) = 4$ and $\text{irr}(\alpha, \mathbb{Q}) = \frac{1}{9}f(x)$.

8. Let $\alpha = \sqrt{2}+i$. Then $\alpha^2 = 2+2\sqrt{2}i-1$ so $\alpha^2-1 = 2\sqrt{2}i$. Squaring again, we obtain $\alpha^4-2\alpha^2+1 = -8$, so $\alpha^4 - 2\alpha^2 + 9 = 0$. Let $f(x) = x^4 - 2x^2 + 9$. One can show that $f(x)$ is irreducible by the technique of Example 23.14. Thus $\deg(\alpha, \mathbb{Q}) = 4$ and $\text{irr}(\alpha, \mathbb{Q}) = f(x)$.

9. We see that i is algebraic over $\mathbb{Q}$ because it is a zero of $x^2 + 1$ in $\mathbb{Q}[x]$; $\deg(i, \mathbb{Q}) = 2$.

10. Let $\alpha = 1 + i$. Then $\alpha - 1 = i$ so $\alpha^2 - 2\alpha + 2 = 0$. Because α is not in $\mathbb{R}$, we see that α is algebraic of degree 2 over $\mathbb{R}$.

11. The text told us that π is transcendental over $\mathbb{Q}$, behaving just like an indeterminant. Thus $\sqrt{\pi}$ is also transcendental over $\mathbb{Q}$. [It is easy to see that if a polynomial expression in $\sqrt{\pi}$ is zero, then a polynomial in π is zero. Namely, starting with $f(\sqrt{\pi}) = 0$, move all odd-degree terms to the right-hand side, factor $\sqrt{\pi}$ out from them, and then square both sides.]

12. Because $\sqrt{\pi} \in \mathbb{R}$, it is algebraic over $\mathbb{R}$ of degree 1. It is a zero of $x - \sqrt{\pi}$ in $\mathbb{R}[x]$.

13. Now $\sqrt{\pi}$ is algebraic over $\mathbb{Q}(\pi)$ of degree 2. It is not in $\mathbb{Q}(\pi)$. Remember that π behaves just like an indeterminant x over $\mathbb{Q}$. Note that $\sqrt{x}$ is not in $\mathbb{Q}(x)$, but it is a zero of $y^2 - x$ in $(\mathbb{Q}(x))[y]$.

14. Now π^2 is transcendental over $\mathbb{Q}$ for the text told us that π is transcendental over $\mathbb{Q}$, and a polynomial expression in π^2 equal to zero and having rational coefficients can be viewed as a polynomial expression in π equal to zero with coefficients in $\mathbb{Q}$ and having all terms of even degree.

15. Now $\pi^2 \in \mathbb{Q}(\pi)$ so it is algebraic over $\mathbb{Q}(\pi)$ of degree 1. It is a zero of $x - \pi^2$ in $(\mathbb{Q}(\pi))[x]$.

16. Now π^2 is algebraic over $\mathbb{Q}(\pi^3)$ of degree 3. It is not in $\mathbb{Q}(\pi^3)$, (note that x^2 is not a polynomial in x^3,) but it is a zero of $x^3 - (\pi^3)^2 = x^3 - \pi^6$ in $(\mathbb{Q}(\pi^3))[x]$.

17. We perform a division.

$$\begin{array}{r|l} & x + (1+\alpha) \\ \hline x - \alpha & x^2 + x + 1 \\ & \underline{x^2 - \alpha x} \\ & \quad (1+\alpha)x \\ & \quad \underline{(1+\alpha)x - \alpha^2 - \alpha} \\ & \qquad\qquad \alpha^2 + \alpha + 1 = 2 \cdot (\alpha + 1) = 0. \end{array}$$

We have $x^2 + x + 1 = (x - \alpha)(x + \alpha + 1)$.

18. **a.** Let $f(x) = x^2 + 1$. Then $f(0) = 1, f(1) = 2$, and $f(-1) = 2$ so $f(x)$ is a cubic with no zeros in $\mathbb{Z}_3$ and thus is irreducible in $\mathbb{Z}_3[x]$.

b.

$+$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
1	1	2	0	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$	α	2α
2	2	0	1	$2+\alpha$	$2+2\alpha$	α	2α	$1+\alpha$	$1+2\alpha$
α	α	$1+\alpha$	$2+\alpha$	2α	0	$1+2\alpha$	1	$2+2\alpha$	2
2α	2α	$1+2\alpha$	$2+2\alpha$	0	α	1	$1+\alpha$	2	$2+\alpha$
$1+\alpha$	$1+\alpha$	$2+\alpha$	α	$1+2\alpha$	1	$2+2\alpha$	2	2α	0
$1+2\alpha$	$1+2\alpha$	$2+2\alpha$	2α	1	$1+\alpha$	2	$2+\alpha$	0	α
$2+\alpha$	$2+\alpha$	α	$1+\alpha$	$2+2\alpha$	2	2α	0	$1+2\alpha$	1
$2+2\alpha$	$2+2\alpha$	2α	$1+2\alpha$	2	$2+\alpha$	0	α	1	$1+\alpha$

$\cdot$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	0	0	0	0	0	0	0	0
1	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
2	0	2	1	2α	α	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	$1+\alpha$
α	0	α	2α	2	1	$2+\alpha$	$1+\alpha$	$2+2\alpha$	$1+2\alpha$
2α	0	2α	α	1	2	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	$2+\alpha$
$1+\alpha$	0	$1+\alpha$	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	2α	2	1	α
$1+2\alpha$	0	$1+2\alpha$	$2+\alpha$	$1+\alpha$	$2+2\alpha$	2	α	2α	1
$2+\alpha$	0	$2+\alpha$	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	1	2α	α	2
$2+2\alpha$	0	$2+2\alpha$	$1+\alpha$	$1+2\alpha$	$2+\alpha$	α	1	2	2α

19. The definition is incorrect. The polynomial must be nonzero and in $F[x]$.

An element α of an extension field E of a field F is **algebraic over** F if and only if α is a zero of some nonzero polynomial in $F[x]$.

20. The definition is incorrect. The polynomial must be nonzero.

An element β of an extension field E of a field F is **transcendental over** F if and only if β is not a zero of any nonzero polynomial in $F[x]$.

21. The definition is incorrect. Only the coefficient of the leading term need be 1.

A **monic polynomial** in $F[x]$ is a nonzero polynomial having 1 as the coefficient in the term of highest degree.

22. The definition is incorrect. The subfields examined must contain F as well as α.

A field E is a **simple extension** of a subfield F if and only if there exists some $\alpha \in E$ such that no proper subfield of E contains both F and α.

23. T T T T F T F T F T

24. **a.** One such field is $\mathbb{Q}(\pi^3)$.

b. One such field is $\mathbb{Q}(e^{10})$.

25. **a.** Let $f(x) = x^3 + x^2 + 1$. Then $f(0) = 1$ and $f(1) = 1$ so $f(x)$ has no zeros in $\mathbb{Z}_2$ and is thus irreducible.

b. The long division uses the relations $\alpha^3 = \alpha^2 + 1$ and $-\alpha^3 - \alpha^2 = -\alpha^2 - 1 - \alpha^2 = -1$.

$$
\begin{array}{r|l}
 & x^2 + (1+\alpha) + (\alpha^2+\alpha) \\
\hline
x-\alpha & x^3 + x^2 + 1 \\
 & \underline{x^3 - \alpha x^2} \\
 & (1+\alpha)x^2 \\
 & \underline{(1+\alpha)x^2 - (\alpha^2+\alpha)x} \\
 & (\alpha^2+\alpha)x + 1 \\
 & \underline{(\alpha^2+\alpha)x - 1} \\
 & 0
\end{array}
$$

Continuing, we try α^2 as a zero of $q(x) = x^2 + (1+\alpha)x + (\alpha^2+\alpha)$. Substituting, we obtain

$$
\begin{aligned}
\alpha^4 + (1+\alpha)\alpha^2 + (\alpha^2+\alpha) &= \alpha(\alpha^2+1) + \alpha^2 + (\alpha^2+1) + (\alpha^2+\alpha) \\
&= (\alpha^2+1) + \alpha + \alpha^2 + (\alpha^2+1) + (\alpha^2+\alpha) \\
&= 2\cdot(\alpha^2+1) + 2\cdot\alpha^2 + 2\cdot\alpha = 0
\end{aligned}
$$

so α^2 is a zero of $q(x)$. We do another long division. This one involves the computation

$$\begin{aligned}\alpha^2(\alpha^2+\alpha+1) &= \alpha\alpha^3+\alpha^3+\alpha^2\\ &= \alpha(\alpha^2+1)+(\alpha^2+1)+\alpha^2\\ &= (\alpha^2+1)+\alpha+(\alpha^2+1)+\alpha^2\\ &= \alpha^2+\alpha.\end{aligned}$$

$$\begin{array}{r|l} & x+(\alpha^2+\alpha+1)\\ \hline x-\alpha^2 & x^2 + (1+\alpha)x+(\alpha^2+\alpha)\\ & \underline{x^2 - \alpha^2 x}\\ & (\alpha^2+\alpha+1)x+(\alpha^2+\alpha)\\ & \underline{(\alpha^2+\alpha+1)x+(\alpha^2+\alpha)}\\ & 0\end{array}$$

Thus in $(\mathbb{Z}_2)[x]$,

$$x^3+x^2+1=(x-\alpha)(x-\alpha^2)[x-(\alpha^2+\alpha+1)].$$

26. The group $\langle \mathbb{Z}_2(\alpha), +\rangle$ is abelian of order 8 with the property that $a+a=0$ for all elements a in the group. Thus the group must be isomorphic to $\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2$. The group $\langle \mathbb{Z}_2(\alpha)^*, \cdot\rangle$ is abelian of order 7, and must be cyclic (both because it has prime order and because it is the multiplicative group of nonzero elements of a finite field) and is isomorphic to $\mathbb{Z}_7$.

27. It is the monic polynomial in $F[x]$ of *minimal* degree having α as a zero.

28. Take an irreducible factor $p(x)$ of $f(x)$, and form the field $E=F[x]/\langle p(x)\rangle$. If we identify each $a\in F$ with the coset $a+\langle p(x)\rangle$ in $F[x]/\langle p(x)\rangle$, then we can view E as an extension field of F. The coset $\alpha = x+\langle p(x)\rangle$ can be viewed as a zero in E of $p(x)$, and hence as a zero in E of $f(x)$.

29. Every element of $F(\beta)$ can be expressed as a quotient of polynomials in β with coefficients in F. Because α is algebraic over $F(\beta)$, there is a polynomial expression in α with coefficients in $F(\beta)$ which is equal to zero. By multiplying this equation by the polynomial in β which is the product of the denominators of the coefficients in this equation, we obtain a polynomial in α equal to zero and having as coefficients polynomials in β. Now a polynomial in α with coefficients that are polynomials in β can be formally rewritten as a polynomial in β with coefficients that are polynomials in α. [Recall that $(F[x])[y]\simeq(F[y])[x]$.] This polynomial expression is still zero, which shows that β is algebraic over $F(\alpha)$.

30. Theorem 29.18 shows that every element of $F(\alpha)$ can be uniquely expressed in the form

$$b_0+b_1\alpha+b_2\alpha^2+\cdots+b_{n-1}\alpha^{n-1}.$$

Because F has q elements, there are q choices for b_0, then q choices for b_1, etc. Thus there are q^n such expressions altogether. The *uniqueness* property shows that different expressions correspond to distinct elements of $F(\alpha)$, which must therefore have q^n elements.

31. **a.** Let $f(x)=x^3+x^2+2$. Then $f(0)=2, f(1)=1$, and $f(-1)=2$ so $f(x)$ has no zeros in $\mathbb{Z}_3$ and thus is irreducible over $\mathbb{Z}_3[x]$.

b. Exercise 30 shows that the field $\mathbb{Z}_3[x]/\langle f(x)\rangle$, which can be viewed as an extension field of $\mathbb{Z}_3$ of degree 3, has $3^3=27$ elements.